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Identification of tomato ripeness levels (*Lycopersicum esculentum* Miil) using android-based digital image processing

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Abstract. The process of identifying tomato ripeness is generally done based on human perception, which is subjective and takes a long time. Color is one indicator of the tomato ripeness level. The color of tomato skin is an important indicator in determining the level of ripeness [1]. The development of image processing methods allows the identification of tomato ripeness levels more accurately, precisely and quickly. Digital image processing refers to processing two-dimensional images using a computer and using digital image processing techniques to obtain information that can be processed by a computer and the results are more precise, fast and automatic. One of the reasons for using Android-based applications as a platform is because it is easily accessible anywhere. Based on the results of the research, the identification of the ripeness level of tomatoes using image processing methods with the C# programming language with Unity 3D software can be implemented to determine the ripeness level of tomatoes. The validation of the chromameter and digital image obtained the coefficient of determination La^*b^* , respectively 0.9979, 0.9999, 0.9978 and the correlation value of La^*b^* is 0.998, 0.999 and 0.998. The designed application can be implemented with RMSE (Root Mean Squared Error) values at La^*b^* of 0.171, 0.530, 0.537.

1. Introduction

One of the stages in processing agricultural products is the selection of products based on their quality. Identification of ripeness level is important in determining product quality. The level of ripeness of tomatoes is usually seen from the skin color. According to [1] the color of tomato skin is an important indicator in determining the level of ripeness. The process of determining the level of ripeness is generally still done manually with visual observations based on human perception. Weaknesses in manual identification are influenced by human subjectivity, resulting in a variety of products, less accurate observations and takes a long time. This is caused by human visual limitations, levels of fatigue and differences in perceptions of fruit quality.

The development of science and technology for digital image processing based on Android makes it possible to determine product quality based on ripeness level automatically so that it can facilitate and improve performance in identifying the ripeness level of tomatoes. Digital image processing is a technique for digitally processing the image of an object which after going through the processing process will obtain information that can be processed by a computer for various needs. Digital image is an optical data recording system in the form of photos, analog in the form of video signals such as images on a television monitor or digital which is directly stored on a magnetic tape [2]. The selection



of an Android-based platform cellular phone is one of the application developments besides being open source, easier to operate and flexible. According to [1], the increasing number of smartphone users makes the development of Android-based applications more and more. This is what makes this application will be developed into an Android-based application so that it will be easier to access on each smartphone. This study uses an Android-based application with Unity 3D software and the C# programming language. Unity 3d software and the C# programming language have the advantage of a simpler program display and easier operation. The aims of this research are analyze image processing methods in determining the level of ripeness of tomatoes based on fruit skin color with the HSV (Hue, Saturation, Value) color model and application application in determining the level of ripeness of tomatoes based on fruit skin color with the C# program language with Android-based Unity 3D software.

Based on the description above, it is necessary to do research using Android-based digital image programming with Unity 3d software and C# programming language with the title "Identification of Tomato Ripeness Level (*Lycopersicum esculentum* Mill) Using Android-Based Digital Image Processing".

2. Materials and methods

Based on the results of the preliminary research, there were 3 classes of ripeness levels based on the skin color of tomatoes, e.g red skin color is ripe tomatoes, yellow skin color is half ripe tomatoes, and green skin color is unripe tomatoes. The tomatoes used in this study came from Alahan Panjang, West Sumatera Province. The range of HSV (Hue, Saturation, Value) used for the ripe category is ($H < 15$, $85 \leq S \leq 90$, $55 \leq V \leq 80$), the HSV (Hue, Saturation, Value) value for the semi-ripe category is ($15 \leq H \leq 30$, $90 \leq S \leq 95$, $50 \leq S \leq 65$) and the unripe category HSV (Hue, Saturation, Value) values are ($H > 30$, $70 \leq S \leq 80$, $25 \leq V \leq 50$).

Image Capture

The process of taking pictures to determine the maturity level of tomatoes using Android with a minimum pixel resolution of the camera used is 8 mp [2]. The maximum distance in taking photos is 25 cm [3]. Input the original image from the folder then cropping the original image on the tomato skin aims to reduce the size of the fruit, after cropping it will resize the cutting image to 32x32 pixels to reduce the computational process [4].

The next process in digital image processing will get RGB values and then convert them to HSV values using the following formula:

$$V = \frac{\max}{225} \times 100 \quad (1)$$

where :

max is the result value of the formula MAX(R,G,B)

$$S = 0 \quad , \text{ If } V=0 \quad (2)$$

$$S = \frac{\max - \min}{\max} \times 100 \quad , \text{ If } V>0 \quad (3)$$

where :

max is the result value of the formula MAX(R,G,B)

min is the result value of the formula MIN(R,G,B)

V is color Value

$$H = 0 \quad \text{If } S=0 \quad (4)$$

$$H=60 \times \frac{G-B}{\Delta} \quad \text{If } \max = R \quad (5)$$

$$H=60 \times \left(2 + \frac{B-R}{\Delta}\right) \quad \text{If } \max = G \quad (6)$$

$$H=60 \times \left(4 + \frac{R-G}{\Delta}\right) \quad \text{If } \max = B \quad (7)$$

$$\text{If } H < 0, \quad H=H+360 \quad (8)$$

$$\Delta = \max - \min$$

The next process in digital image processing is to get RGB values and then convert them to L* a* b* values. The Lab color space consists of L*, a*, and b*. It is necessary to convert from the RGB color space to the XYZ color space first before going to the Lab [5]. The RGB to XYZ conversion formula can be seen in Equation 9.

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} 0,412453 & 0,357580 & 0,180423 \\ 0,212671 & 0,715160 & 0,072169 \\ 0,119193 & 0,119193 & 0,950227 \end{pmatrix} \begin{pmatrix} R \\ G \\ B \end{pmatrix} \quad (9)$$

After conversion, the Lab color space value can be calculated. The L* color formula can be seen in Equation 10. The a* value formula can be seen in Equation 11 and the b* formula can be seen in Equation 12 [5]

$$L = \begin{cases} 116 \left(\frac{X}{Y_n} \right)^{1/3} - 16 & \text{if } \left(\frac{X}{Y_n} \right) > 0,008856 \\ 903,3 \left(\frac{X}{Y_n} \right)^{1/3} - & \text{if } \left(\frac{X}{Y_n} \right) \leq 0,008856 \end{cases} \quad (10)$$

$$a^* = 500 \left[f\left(\frac{X}{Y_n}\right) - f\left(\frac{Y}{Y_n}\right) \right] \quad (11)$$

$$b^* = 200 \left[f\left(\frac{X}{Y_n}\right) - f\left(\frac{Z}{Z_n}\right) \right] \quad (12)$$

The values of X_n, Y_n, and Z_n have values that can be seen in Equation 13. The formula for the function f(u) can be seen in Equation 14.

$$\begin{bmatrix} X_n & Y_n & Z_n \end{bmatrix} = \begin{bmatrix} 0,950490 & 1,000000 & 0,088754 \end{bmatrix} \quad (13)$$

$$f(u) = \begin{cases} u^{1/3} & \text{if } u > 0,008856 \\ 7,787u + \frac{16}{116} & \text{if } u \leq 0,008856 \end{cases} \quad (14)$$

Description: R = value of red color
G = Value of green color
B = Value of blue color

System planning

The design is carried out to describe the input and output interfaces of the application to be built. The application interface is also designed to facilitate the development of applications that are built. The design made about the user interface is a very important part in running this application, because the first thing that will be seen when the user opens the application to be run is the user interface. The following is the user interface of the application that determines the level of maturity of tomatoes based on Android, which can be seen in Figure 1.



Figure 1. Main Menu Display Design and Main Button Display

In Figure 1 there are several main user interface menus that are used from making applications to determine the level of maturity of tomatoes. Some of the main menus used such as images or text will be created containing text and application icons for tomato ripeness, images or text will be created containing the name of the application's main menu, images or text will be created containing text and images regarding the tomato application, images or the text that will be created containing the footer text that will display the making of the tomato application. The button used is the camera button used to take pictures through the android device camera, the gallery button is used to take pictures through the gallery on the android device, the sample button is used to view the stored tomato samples, the method button is used to view the description of the method used to determine maturity tomatoes.

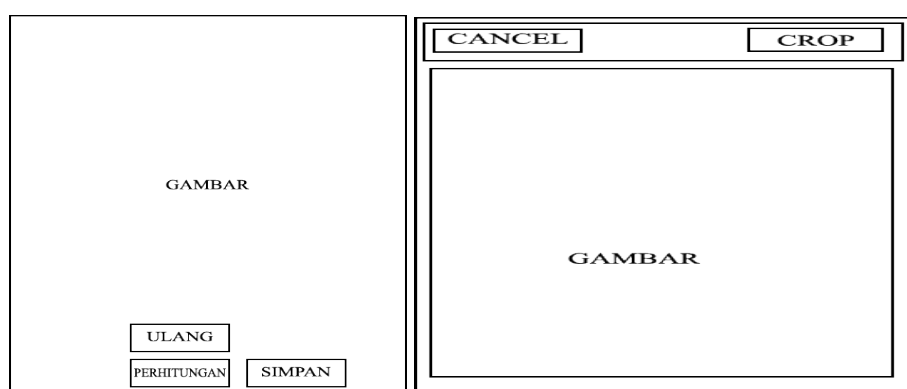


Figure 2. Display of Process Menu and Image Processing Button

From the display in Figure 2, it can be seen that several menus and buttons are used, such as the image menu showing images of tomatoes that have been taken, the repeat button is used to repeat the tomato image, the calculation button is used to view the results of calculations that have been processed, the save button is used to store the results of the tomatoes. has been processed. Some of the display buttons used are the cancel button used to cancel the tomato photos that have been taken, the crop

button to crop the tomato images that have been taken, images to display the tomato images that have been taken.

NO	GAMBAR	R	G	B	H	HASIL

METODE	

Figure 3. Display of Image Processing Results and Methods

Display of Figure 3 can be seen several views of the processing results such as No displays the number of tomatoes that have been stored, the Image displays images of tomatoes that have been saved, R displays the R value that has been stored, G displays the G value that has been stored, B displays the B value that has been stored. stored, H displays the value of H that has been stored, The results display the results of the ripeness level of tomatoes that have been stored. The method menu that serves to display a description of the method used in the application determines the level of maturity of tomatoes. The output of this research is an android application to determine the maturity level of tomatoes. The research flow chart can be seen in Figure 4.

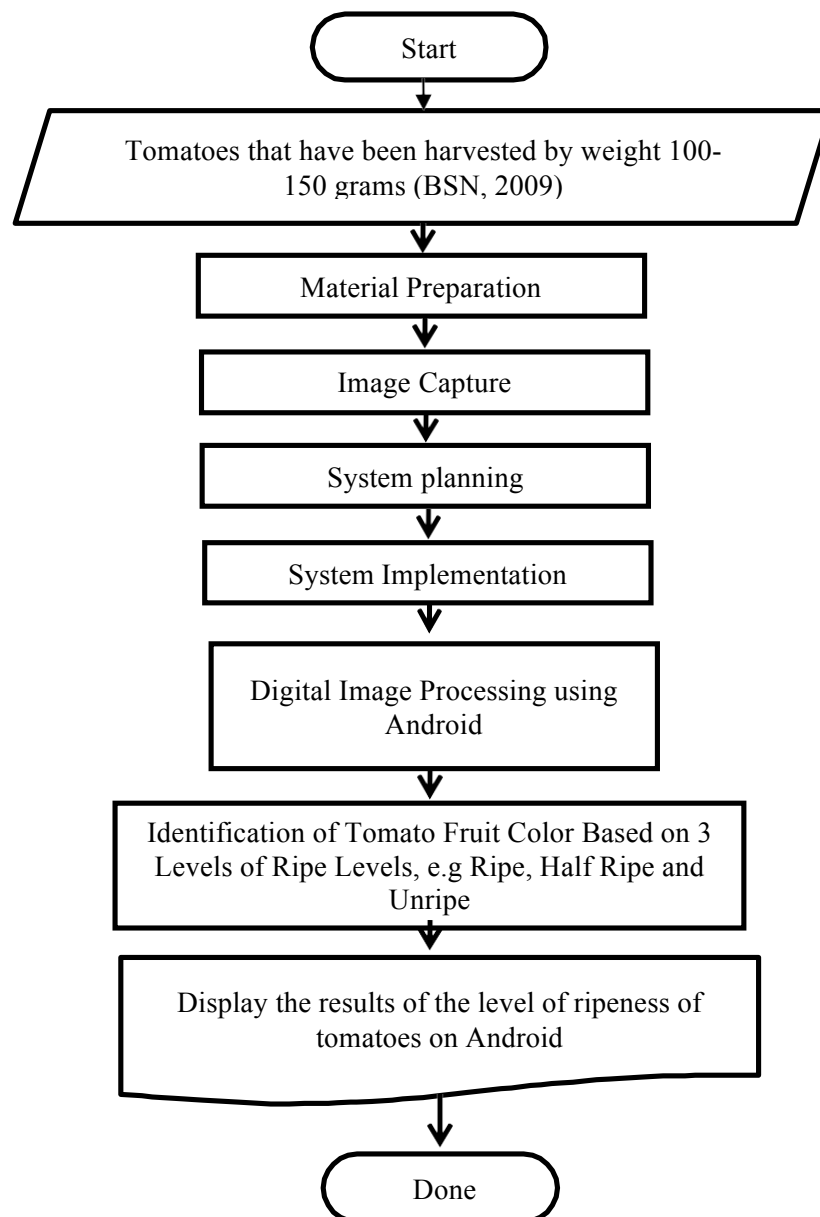


Figure 4. Flowchart of Determination of Ripeness in Tomato Fruit

3. Results and discussion

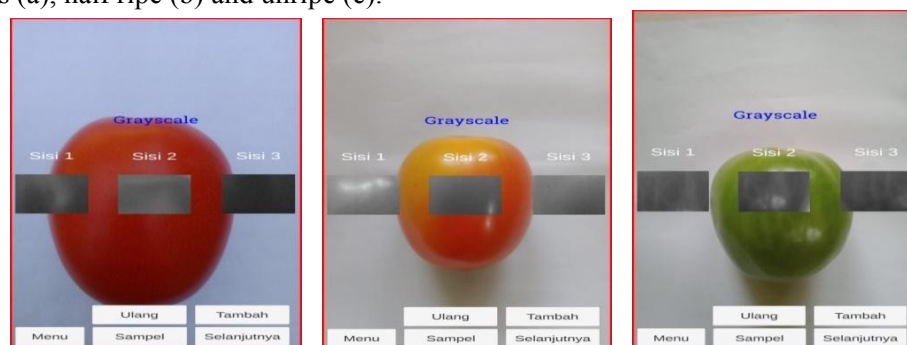
3.1. Program Interface Design

Program interface design is carried out to describe the input and output interfaces of the applications built. The following are some of the features available in the application to determine the maturity level of tomatoes. The splash screen display is the initial display of the application in identifying the maturity level of tomatoes before the application displays the home page. The home view is the main view of the application, where on the home screen there are several main menus, namely the camera menu, gallery menu, sample menu, method menu, about menu and exit menu. The following display splash screen and home can be seen in Figure 5.



Figure 5. Splash Screen and Home Tampilan

The initial stage for the process of identifying the level of maturity with image processing is taking photos of the tomatoes, then cropping the 3 sides of the tomatoes, e.g. the front, the middle and the back. Cropping on tomato skin serves to reduce the size of the tomato fruit, after cropping it will resize the cutting image to 32x32 pixels to reduce the computational process [4]. After cropping the tomatoes, the next step is to convert the image to a grayscale image. Grayscale image is a digital image that only has one channel value in each pixel to show the color intensity value. Change the color of the grayscale image which aims to identify the presence of objects to gray scale. The following Figure 6 shows the process of changing the color of the grayscale image on the three sides of ripe tomatoes (a), half ripe (b) and unripe (c).



(a). Ripe

(b). Half Ripe

(c). Unripe

Figure 6. Color Image of Grayscale on Tomato Fruit

After the grayscale process on the tomatoes, then the grayscale process to RGB, then changes the color from RGB to HSV of each image pixel. The color change of HSV on the three sides of the tomato fruit skin aims to get a color that matches the color captured by human sensing [6]. The color change can be seen in Figure 7

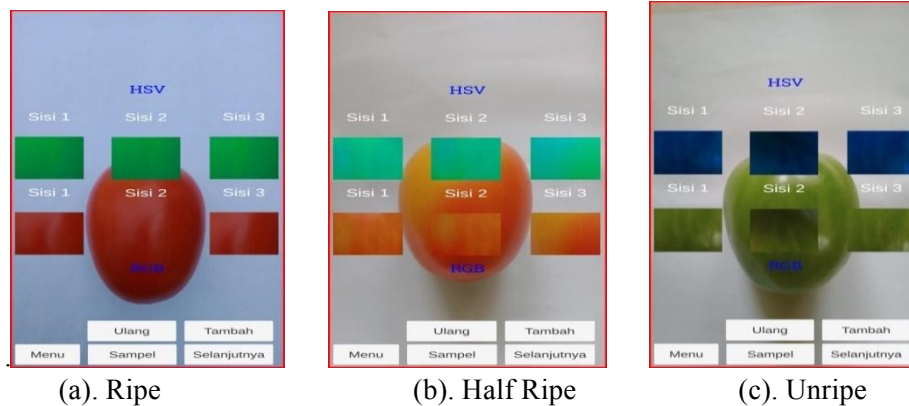
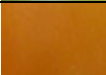


Figure 7. RGB to HSV Color Conversion Results

After the process of changing the color from RGB to HSV, then the value will be converted from RGB to HSV so that the average HSV (Hue, Saturation, Value) value is obtained. After obtaining the average HSV value of each tomato fruit object, the identification of tomatoes according to the stage of maturity level is carried out in Table 4. The results of the conversion of RGB to L a*b* and RGB to HSV values can be seen in Table 1.

Table 1. Result of Conversion of RGB to La*b* and RGB to HSV Values

No	Convert RGB Value to Lab			Convert RGB Value to HSV				Result
	Sample	L	a*	b*	H	S	V	
1								
		47,38	43,73	26,81	3,8	83,4	57,8	Ripe
2.								
		44,55	-20,71	40,33	72,5	74,9	25,4	Unripe
3								
		48,75	11,42	41,51	29,3	91,8	63,3	Half Ripe

Description: Value H (Hue), Value S (Saturation), Value V (Value)
: R Value (Red), G Value (Green), B Value (Blue)

3.2. Test result

Tests were carried out to identify the level of maturity of tomatoes which were carried out manually and based on android. The test samples used were 20 fruit samples. The results of the identification of the level of maturity in tomatoes manually and on an android basis can be seen in Table 2.

Table 2. Comparison Results of Tomato Fruit Ripening Levels Manually and Android-Based

Sample	Results of Identification of Tomato Ripe Level		
	Manually	Android-Base	Information
1	Ripe	Ripe	Suitable
2	Unripe	Unripe	Suitable
3	Unripe	Unripe	Suitable
4	Half Ripe	Half Ripe	Suitable
5	Unripe	Unripe	Suitable
6	Half Ripe	Half Ripe	Suitable
7	Ripe	Ripe	Suitable
8	Half Ripe	Half Ripe	Suitable
9	Unripe	Unripe	Suitable
10	Ripe	Ripe	Suitable
11	Unripe	Unripe	Suitable
12	Half Ripe	Half Ripe	Suitable
13	Ripe	Ripe	Not Suitable
14	Ripe	Ripe	Suitable
15	Unripe	Unripe	Suitable
16	Half Ripe	Half Ripe	Suitable
17	Unripe	Unripe	Suitable
18	Unripe	Unripe	Suitable
19	Half Ripe	Half Ripe	Suitable
20	Ripe	Ripe	Suitable

Based on Table 2, the results of the manual and android-based test results for the system for identification of the Ripe level of tomato fruit, it was found that 19 samples were suitable and there was 1 sample that was not suitable from 20 samples of tomatoes.

3.3. Correlation of color measurement results with a chromameter and digital image.

The results of measuring color chromameter and digital images are based on Lab values. The L value means that it tends to have acrobatic colors of white, gray, and black. The a^* value indicates the color tends to be red and the b^* value indicates the yellow tends to be [1]. Correlation is carried out to see the L a^*b^* relationship between the chromameter and digital image. The results of color measurements with a chromameter and digital image can be seen in Figure 8, it can be seen that the coefficient of determination of the Lab value obtained from the measurement results of the chromameter and digital image is $L = 0.997$, $a^* = 0.999$, $b^* = 0.989$. The correlation value of L a^*b^* is 0.998, 0.999 and 0.989, respectively. Measurement of L a^*b^* color with a chromameter and digital images has a very strong correlation. This is evidenced by the correlation value ($r > 0.800$).

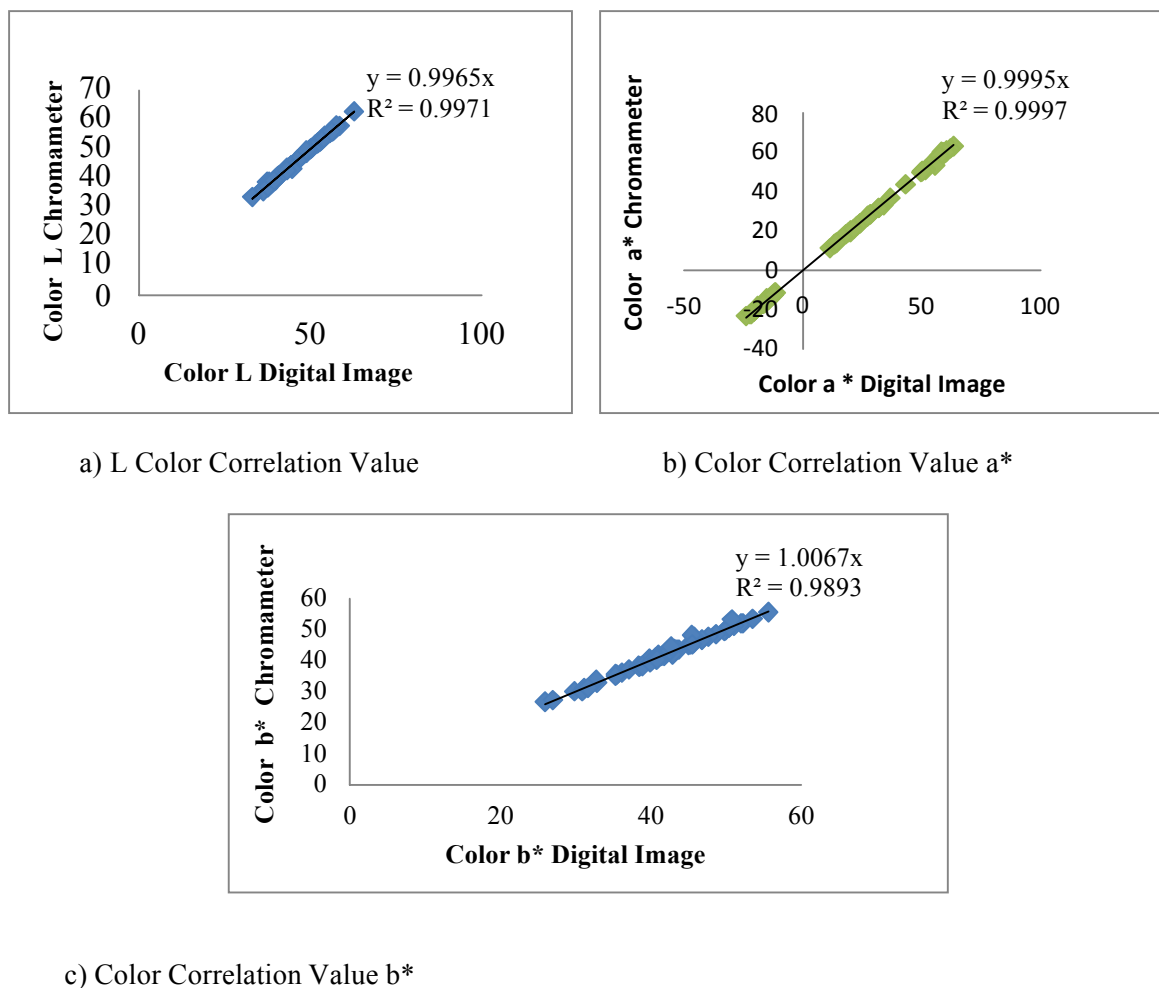


Figure 8. Correlation of La^*b^* Color with Chromameter and Digital Image

The RMSE (Root Mean Squared Error) value obtained at $L^*a^*b^*$ of 0.171, 0.530, 0.537 indicates that the resulting model has a small error rate, so that the designed application can be implemented.

4. Conclusions

Identification of the ripeness level of tomatoes based on the color of the fruit skin with the C# programming language with the Android-based Unity 3D software can be implemented on the Ripean level of tomatoes. Correlation values between $L^*a^*b^*$ chromameter and digital images were 0.998, 0.999 and 0.998. The RMSE (Root Mean Squared Error) values were obtained at $L^*a^*b^*$ of 0.171, 0.530, 0.537. The designed application can be implemented to identify the level of ripeness of tomatoes using Android.

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